PROJECT AVATAR

Agricultural Vision and Automated Technology for Adaptive Response

Off-Grid, IoT-Enabled Smart Agricultural Management and Protection System

A B.Sc. project submitted in partial fulfilment of the requirements for the award of the Bachelor of Science degree in Computer Science.

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Statement of the problem

Many small farms still depend on the farmer being physically present.

- Farm conditions may change between visits.
- Water may be applied without current soil information.
- Fire or unusual movement may be noticed too late.
- Separate devices need separate power supplies and controls.
- Grid electricity and internet access may be unreliable.

The problem is the absence of one small, off-grid system that can monitor the farm and coordinate these responses.

Past solutions and their limitations

Previous work	What it addressed	Limitation
García et al. (2020)	Reviewed sensors and IoT for irrigation.	No integrated management and protection prototype.
Vallejo-Gómez et al. (2023)	Reviewed smart-irrigation systems.	Focused mainly on irrigation.
Pereira et al. (2023)	Built ESP32 irrigation with a dashboard.	No off-grid power or fire response.
Udurume et al. (2025)	Developed networked fire monitoring.	Fire monitoring was separate from farm management.

The gap is integration: previous studies demonstrate useful parts, but not all the selected functions in one off-grid controller.

Our proposed system

Project AVATAR combines farm monitoring, routine control and protection in one ESP32-based prototype.

It monitors

- Soil pH, EC and NPK
- Soil moisture; air temperature and humidity
- Light, gas, movement and fire state
- Data freshness and pump activity
- Future remote child-node readings

It controls

- One bounded irrigation-pump response
- Servo-assisted fire-pump response
- Safe local control without internet
- ThingSpeak data history
- Farmer dashboard and status

System overview and scope

1

Sensors measure the farm.

→

2

The ESP32 checks and stores readings.

→

3

Bounded local protection responds.

→

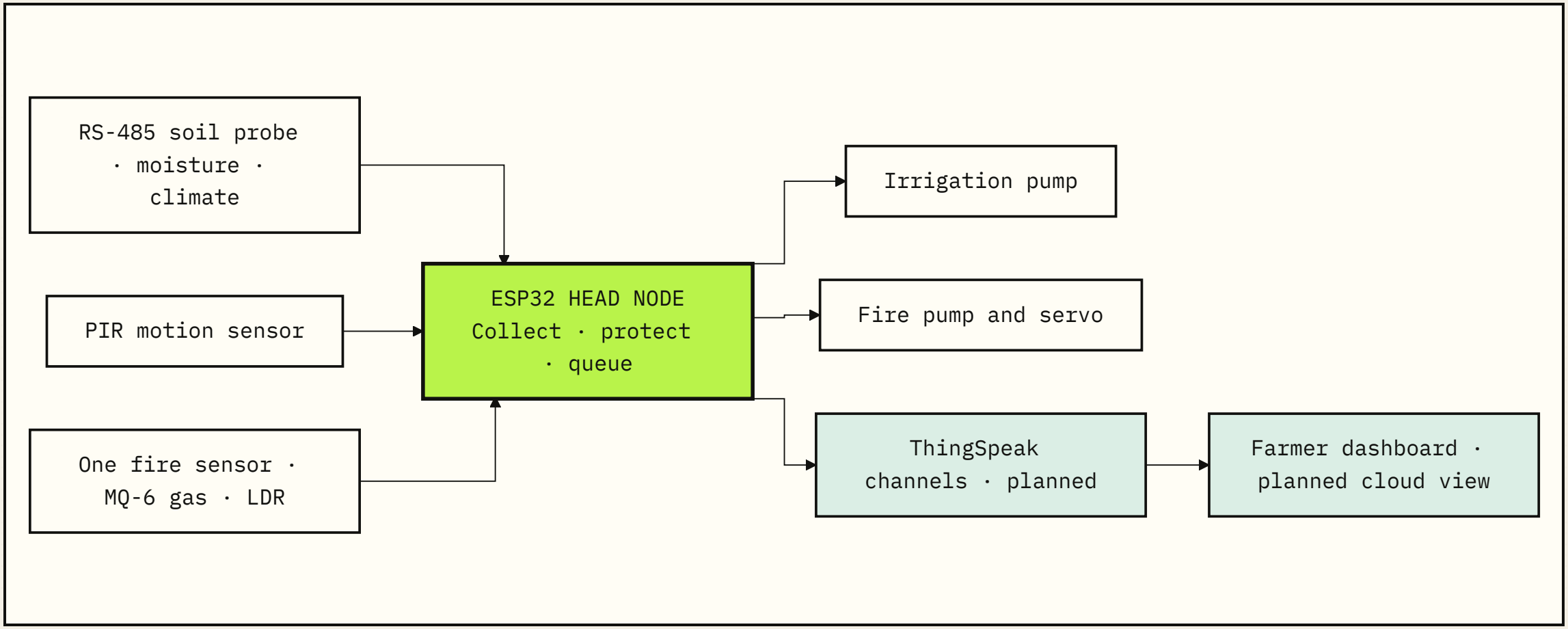
4

ThingSpeak and dashboard show history.

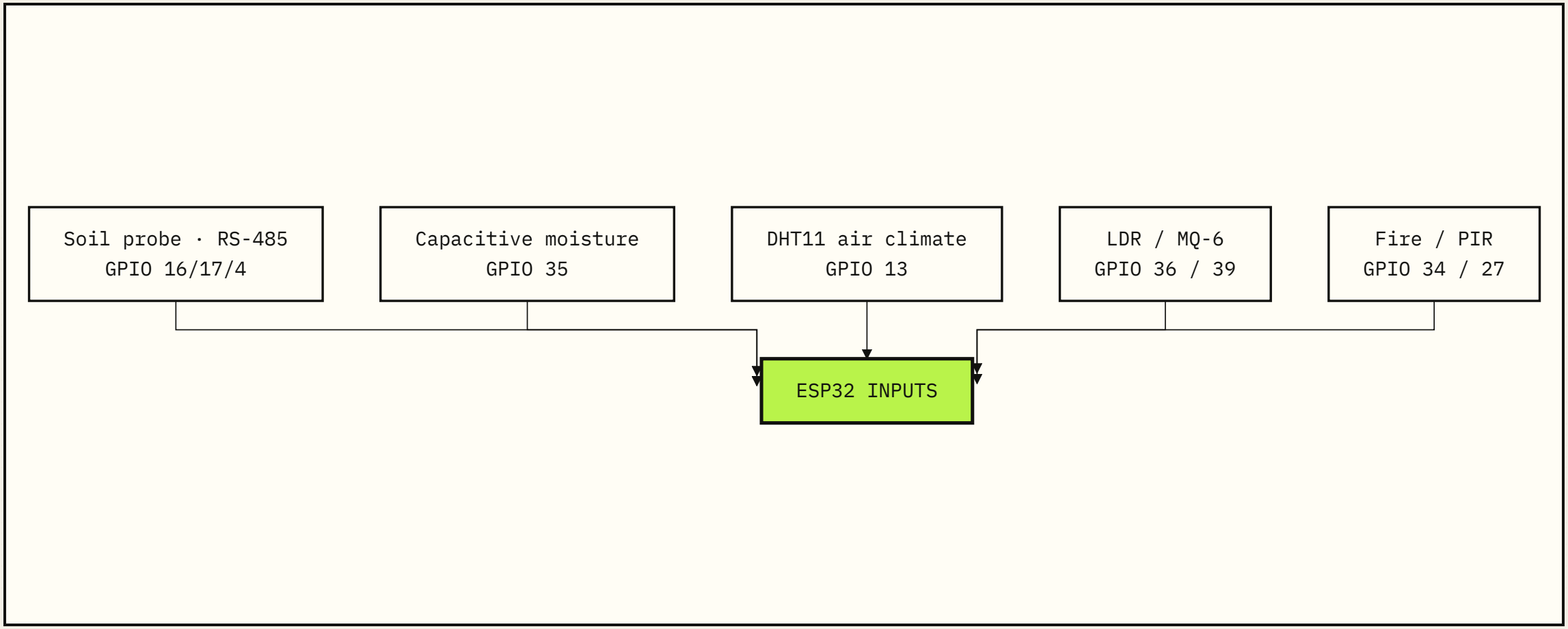
Scope limitation

This is a controlled 2 x 3 ft wired prototype. It demonstrates one irrigation area and one fire response; it does not claim multi-zone irrigation, commercial-farm deployment, crop-yield measurement, or professional fire suppression. Future child nodes will communicate wirelessly after the wired head-node is proven.

Technical drawing: system architecture

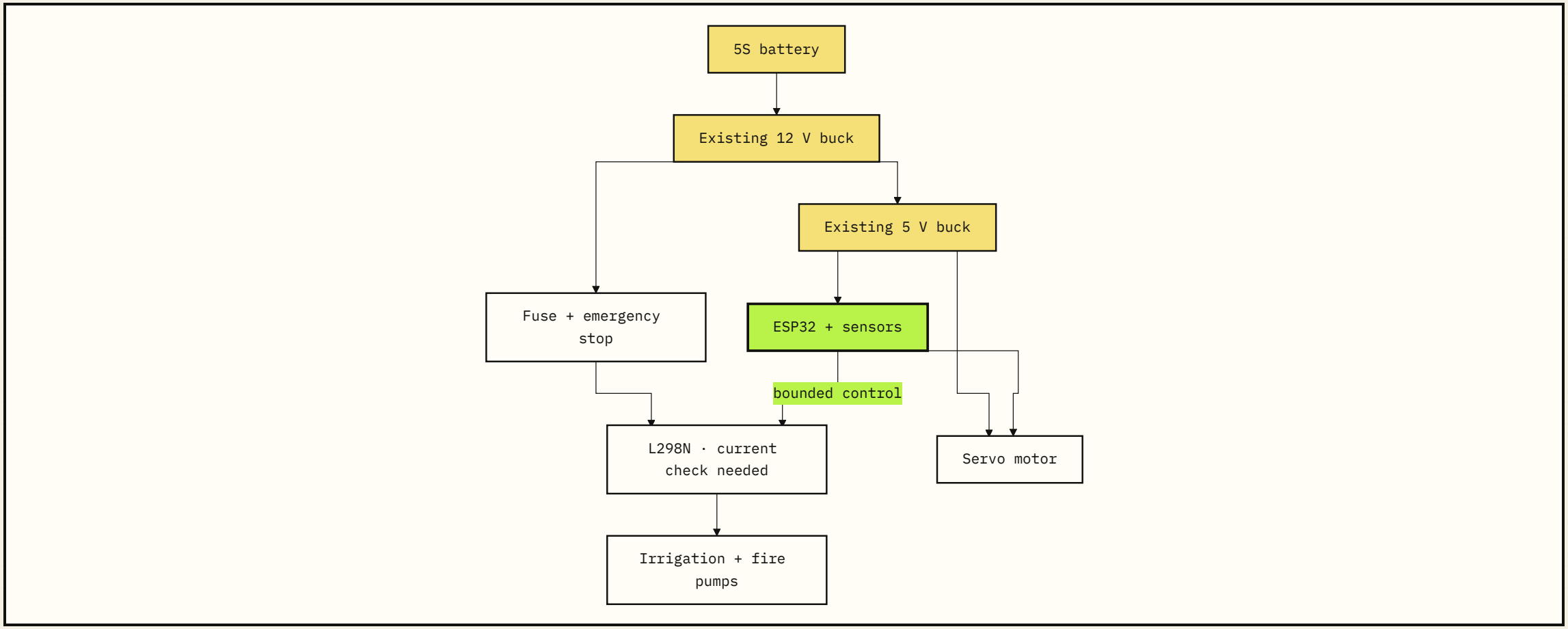


Technical drawing: sensor inputs



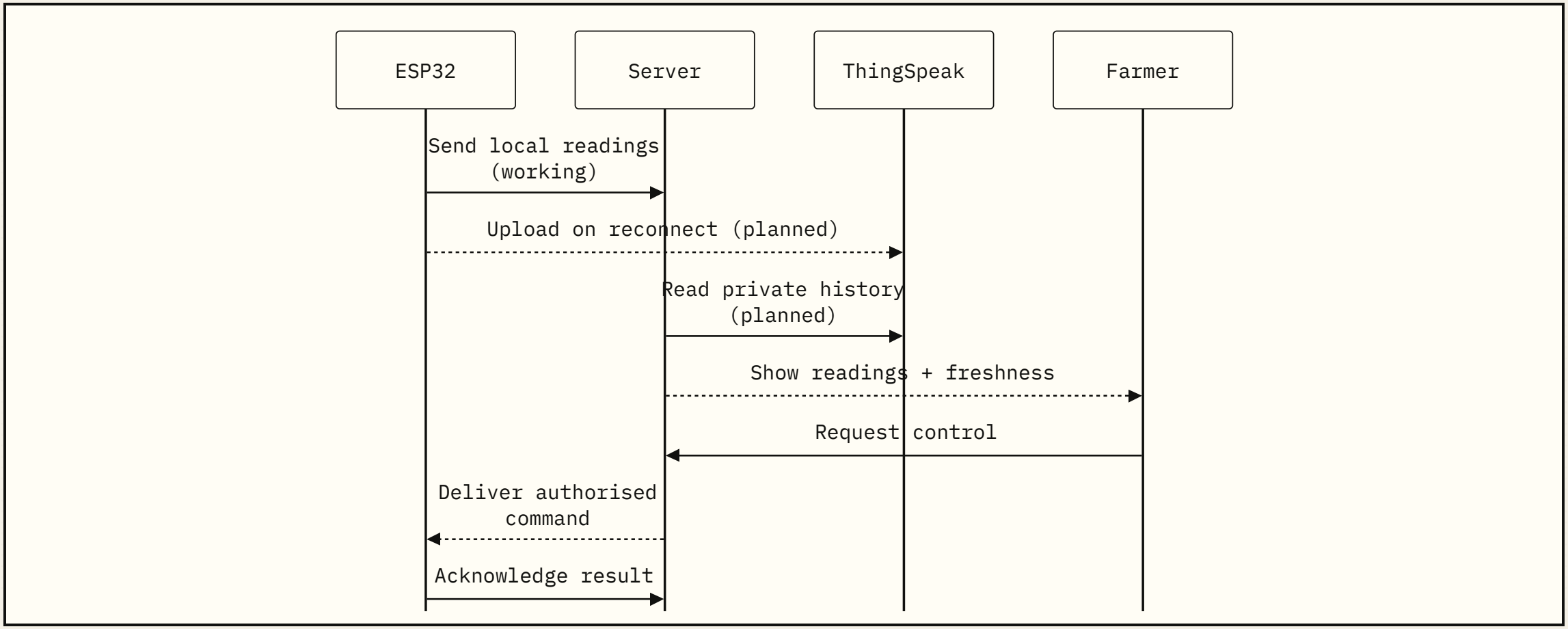
All inputs share regulated power and ground. Analogue lines must stay within 3.3 V and be calibrated; the MAX485 receive line needs a verified 3.3 V interface. The soil probe's register map is not yet confirmed.

Technical drawing: outputs and power



The 5S → 12 V → 5 V buck chain is already in place. Solar charging is deferred until the correct 5S BMS and charger are available. Pump output stays disabled until current and wiring checks pass.

Technical drawing: software communication



Local telemetry to the Node server works now. ThingSpeak upload and the authorised remote data path are planned. Immediate safety decisions remain on the ESP32.

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Thank you.

Questions and comments are welcome.

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